

Appendix: Formal Conjectures for Future Investigation

The following conjectures extend the HFG framework beyond the Standard Model flavor sector. They are stated precisely enough to be falsifiable but are not claimed as theorems. All should be treated as working hypotheses pending further computation and experimental input.

C1. Dark sector arithmetic

Conjecture 1 (Dark Sector Arithmetic). *Let $\mathcal{P}_{\text{visible}} = \{2, 3, 5, 7, 11, 29, 47, 199, \dots\}$ be the primes appearing in the covering towers of the visible sector manifolds M_{PMNS} and M_{CKM} (prime Lucas numbers together with the ancestor torsion 5). Any prime $p \notin \mathcal{P}_{\text{visible}}$ that appears as torsion in the covering tower of a cusped hyperbolic 3-manifold is a candidate for a dark sector particle.*

Specifically, the primes $\{13, 17, 19, 23, 31, 41, 43, \dots\}$ observed in the control scan of m004, m007, m009, m010, m015 (Section ??) identify candidate dark sector manifolds. The mass of a dark fermion associated with prime p is predicted to satisfy:

$$\frac{m_{\text{dark}}}{m_e} = \phi^{q/4},$$

where $q/4$ is a quarter-integer with $q \equiv 0 \pmod{p}$ and $q/4$ is not a quarter-Lucas index.

Mass predictions for candidate primes.

Prime p	$q = p$	$m \approx m_e \cdot \phi^{p/4}$	Note
13	13	$0.511 \times \phi^{3.25} \approx 1.85 \text{ MeV}$	sterile neutrino range
17	17	$0.511 \times \phi^{4.25} \approx 4.8 \text{ MeV}$	dark photon range
19	19	$0.511 \times \phi^{4.75} \approx 6.4 \text{ MeV}$	light dark matter
31	31	$0.511 \times \phi^{7.75} \approx 39 \text{ MeV}$	possible dark sector mediator
41	41	$0.511 \times \phi^{10.25} \approx 580 \text{ MeV}$	sub-GeV dark matter
43	43	$0.511 \times \phi^{10.75} \approx 940 \text{ MeV}$	near proton mass

Consistency check. The 17 MeV dark photon anomaly (claimed in ^8Be nuclear transitions) implies $q/4 \approx \log(17/0.511)/\log \phi \approx 7.28$, so $q \approx 29.1$. Since $29 = L_7$ is a prime Lucas number, this anomaly would lie in the *visible* sector, not the dark sector, in the HFG framework. The prediction is therefore that the 17 MeV anomaly is not a dark sector effect.

Falsifiability. A discovered dark matter particle with mass m_{dark} can be tested by computing $q/4 = \log(m_{\text{dark}}/m_e)/\log \phi$ and checking whether $q/4$ is a quarter-Lucas index or not. If quarter-Lucas: visible sector (contradicts the conjecture). If non-Lucas quarter-integer: consistent with dark sector prediction.

C2. Cosmological unfolding

Conjecture 2 (Cosmological Unfolding). *The transition from matter-dominated to dark-energy-dominated cosmological expansion corresponds to the transition from the sparse spectral tier to the universal spectral tier in the geodesic length spectrum of Dehn fillings of m003 and m006.*

Formally: let $n^(\text{m003}) \approx 21$ and $n^*(\text{m006}) \approx 18$ be the phase transition thresholds. The matter-energy equality redshift $z_{\text{eq}} \approx 0.3$ corresponds to the spectral threshold $n^* \approx 18\text{--}21$ (equivalently, $\log n^* \approx 2.9\text{--}3.0$ in units of $\log \phi$, which gives $k^* = \log n^*/\log \phi \approx 6.0\text{--}6.4$, i.e., $L_{k^*} \approx 18\text{--}23$).*

The dark energy fraction $\Omega_\Lambda \approx 0.70$ is consistent with the average universal-tier occurrence at threshold: $\bar{f}(n^*) = (65.6\% + 81.8\%)/2 \approx 73.7\%$ (see Table ??).

Remark 1. The 73.7% vs 70% agreement is numerically suggestive but currently coincidental — no mechanism has been identified that would enforce this equality. We record it as an observation that motivates further investigation.

Hubble tension connection. The threshold discrepancy between sectors — $n^*(m006) = 18$ (quark) vs $n^*(m003) = 21$ (lepton) — implies that measurements of the Hubble parameter H_0 dominated by early-universe (CMB, quark sector) physics should systematically differ from late-universe (local distance ladder, lepton sector) measurements. The observed Hubble tension $H_0^{\text{early}}/H_0^{\text{late}} \approx 67.4/73.0 \approx 0.923$ is in the same direction as the threshold ratio $n^*(m003)/n^*(m006) = 21/18 \approx 1.17$ (inverted: lepton threshold is higher, so lepton-dominated measurements give lower H_0).

This is not a derivation. It is a structural observation that the two sectors have different spectral thresholds, and that this difference is in the correct direction to explain the Hubble tension qualitatively.

Dehn filling flow model. The cosmological expansion can be modeled as a flow on Dehn filling parameter space:

$$(p(t), q(t)) \longrightarrow (-2, 3) \text{ and } (-5, 2) \quad \text{as } t \rightarrow t_{\text{now}}.$$

Early universe ($|p| + |q|$ large): near-cusp regime, universal spectral tier dominates, no slope-specific structure (symmetric phase, no distinct particles). Late universe: specific slopes reached, sparse tier emerges, Standard Model particle content appears.

The Hubble parameter in this model is:

$$H(t) \propto \frac{d}{dt} [\log \text{vol}(M(p(t), q(t)))] ,$$

which decreases as $|p| + |q|$ decreases (volume decreases toward the filled manifold). The dark energy acceleration corresponds to the slowdown in volume change as the filling approaches the target slopes $(-2, 3)$ and $(-5, 2)$.

C3. Information-geometry correspondence

Conjecture 3 (Ihara–Selberg Bridge). *The geodesic length spectrum of M_{PMNS} is the Riemannian analog of the Ihara zeta function of a binary tree of depth $k_{\text{max}} = 9$ (the covering tower degree). Specifically, the information capacity of a binary tree of depth k is $\log_2 F_k$ (where F_k is the k -th Fibonacci number), and $F_k \sim \phi^k / \sqrt{5}$, so the capacity grows as $k \log_2 \phi$. The fundamental geodesic unit $\log \phi$ of M_{PMNS} and the information unit $\log_2 \phi$ of the Fibonacci binary tree are the same object in different bases.*

Connection to error-correcting codes. Reed-Solomon codes (used in hard drive ECC) operate over $\text{GF}(2^k)$ — finite fields of characteristic 2. The prime $2 = L_0$ is the first prime Lucas number and the base of binary arithmetic. The covering tower of M_{PMNS} introduces $L_0 = 2$ at degree 3 (Table ??). This suggests a connection between the degree-3 cover (the first cover introducing the prime 2) and the 3-bit encoding level of binary codes, though the mechanism is not yet understood.